

ILO INNOCENT EZEKIEL

CYBER SECURITY ASSIGNMENT 3

ATTACKER MACHINE: KALI

TARGET MACHINE: METASPLOITABLE 2

Nmap -sS was used to perform a basic TCP SYN scan on the target to check which ports are open without completing a full connection.

FINDINGS: Host is up, tcp ports open with services like ftp, ssh, telnet, smtp, domain, http, and more running. Only the open ports were shown.

```
(kali㉿kali)-[~]  
$ nmap -sS 10.80.80.100  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 07:56 EDT  
Nmap scan report for 10.80.80.100  
Host is up (0.0035s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:C0:C2:B7 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 13.40 seconds
```

Nmap -sV was used to know what services, which version of services running in the target machine.

FINDINGS: Only opened tcp ports shown running on different services with their versions such as port 21/tcp open with ftp service and version vsftpd 2.3.4 running on it.

```
L$ nmap -sV 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 08:00 EDT
Nmap scan report for 10.80.80.100
Host is up (0.00067s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:C0:C2:B7 (VMware)
```

Using of Nmap Scripting Engine:

These goes beyond scanning but also detecting vulnerabilities, gather detailed system information, perform advanced service checks and automate common security tasks.

- i. Nmap --script=vuln to run all scripts in the vulnerability category

FINDINGS: Ports, state, and services running on the target were shown including vulnerabilities.

Port 21/tcp is open with ftp service and vstpd version 2.3.4 running on it which is vulnerable and can be exploited through the backdoor. It has a critical severity , Remote

shell access, full system compromise, and a vulnerability of CVE-2011-2523. No authentication required

```
└─$ nmap --script=vuln 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 08:06 EDT
Nmap scan report for 10.80.80.100
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs: BID:48539 CVE:CVE-2011-2523
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         https://www.securityfocus.com/bid/48539
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|_
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
| smtp-vuln-cve2010-4344:
|_ The SMTP server is not Exim: NOT VULNERABLE
53/tcp    open  domain
80/tcp    open  http
|_ http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
| http-sql-injection:
|   Possible sqlmap queries:
|     http://10.80.80.100:80/dav/?C=S%3B0%3DA%27%200R%20sqlspider
|     http://10.80.80.100:80/dav/?C=D%3B0%3DA%27%200R%20sqlspider
|     http://10.80.80.100:80/dav/?C=M%3B0%3DA%27%200R%20sqlspider
|     http://10.80.80.100:80/dav/?C=N%3B0%3DD%27%200R%20sqlspider
|     http://10.80.80.100:80/mutillidae/?page=view-someones-blog.php%27%200R%20sqlspider
|     http://10.80.80.100:80/mutillidae/index.php?page=user-poll.php%27%200R%20sqlspider
|     http://10.80.80.100:80/mutillidae/index.php?page=source-viewer.php%27%200R%20sqlspider
|     http://10.80.80.100:80/mutillidae/index.php?page=usage-instructions.php%27%200R%20sqlspider
```

- ii. Nmap -Sv --script=http-tittle -p was used to scan port 80, detect service version and get the web tittle.

FINDINGS: Port 80/tcp is open running of http, version Apache httpd 2.2.8 (ubuntu).

http-tittle: metasploitable 2 -linux.

```
(kali@kali)-[~]
$ nmap -sV --script=http-title -p 80 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:00 EDT
Nmap scan report for 10.80.80.100
Host is up (0.00054s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
MAC Address: 00:0C:29:C0:C2:B7 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.90 seconds
```

- iii. Nmap -sV --script=vuln was used to run multiple vulnerability checks on the target machine.

```
(kali@kali)-[~]
$ nmap -sV --script=vuln 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 08:31 EDT
Nmap scan report for 10.80.80.100
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ ftp-vsftpd-backdoor:
|_ VULNERABLE:
|_ vsFTPD version 2.3.4 backdoor
|_ State: VULNERABLE (Exploitable)
|_ IDs: BID:48539 CVE:CVE-2011-2523
|_ vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|_ Disclosure date: 2011-07-03
|_ Exploit results:
|_ Shell command: id
|_ Results: uid=0(root) gid=0(root)
|_ References:
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|_ https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|_ http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|_ https://www.securityfocus.com/bid/48539
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
|_ smtp-vuln-cve2010-4344:
|_ The SMTP server is not Exim: NOT VULNERABLE
|_ sslv2-drown:
|_ ciphers:
|_ SSL2_DES_64_CBC_WITH_MD5
|_ SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ SSL2_RC4_128_EXPORT40_WITH_MD5
|_ SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ SSL2_RC2_128_CBC_WITH_MD5
|_ SSL2_RC4_128_WITH_MD5
|_ vulns:
|_ CVE-2016-0800:
|_ title: OpenSSL: Cross-protocol attack on TLS using SSLv2 (DROWN)
|_ state: VULNERABLE
```

- iv. Nmap --script=auth was used to check weak or missing authentication, default credentials, authentication methods supported by services, misconfigured login systems.

FINDINGS: Port 21/tcp is open allowing anonymous login.

Port 22/tcp is open running ssh service and supporting password and publickey authentication.

```
(kali㉿kali)-[~]
└─$ nmap --script=auth 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:10 EDT
Nmap scan report for 10.80.80.100
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
| ssh-publickey-acceptance:
|_ Accepted Public Keys: No public keys accepted
| ssh-auth-methods:
|_ Supported authentication methods:
|   publickey
|   password
23/tcp    open  telnet
25/tcp    open  smtp
| smtp-enum-users:
|_ Method RCPT returned a unhandled status code.
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
```

```

(kali㉿kali)-[~]
└─$ nmap --script=http-auth -p 80 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:15 EDT
Nmap scan report for 10.80.80.100
Host is up (0.0010s latency).

PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 00:0C:29:C0:C2:B7 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.58 seconds

```

Using Nmap `--script=vulners` to identify known vulnerabilities (CVE and CVSS)

Full CVE scan on HTTP port 80

FINDINGS: Port 80/tcp is open running http on version Apache httpd 2.2.8 ubuntu.

http-server-header: Apache/2.2.8 ubuntu.

```

(kali㉿kali)-[~]
└─$ nmap -sV --script=vulners -p 80 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:21 EDT
Nmap scan report for 10.80.80.100
Host is up (0.00060s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
MAC Address: 00:0C:29:C0:C2:B7 (VMware)

```

```

(kali㉿kali)-[~]
$ nmap -sV --script vulners 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:26 EDT
Stats: 0:15:31 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 93.76% done; ETC: 09:43 (0:00:59 remaining)
Nmap scan report for 10.80.80.100
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|   program version   port/proto  service
|   100000   2             111/tcp    rpcbind
|   100000   2             111/udp    rpcbind
|   100003   2,3,4         2049/tcp   nfs
|   100003   2,3,4         2049/udp   nfs
|   100005   1,2,3         42373/tcp  mountd
|   100005   1,2,3         56098/udp  mountd
|   100021   1,3,4         34206/udp  nlockmgr
|   100021   1,3,4         44137/tcp  nlockmgr
|   100024   1             56579/udp  status
|_  100024   1             60810/tcp  status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd

```


Use of Nmap -sV --script vulners --script-args mincvss=7.0 to detect service versions, search for known vulnerabilities affecting those versions, display only vulnerabilities with a severity score of 7.0 or above.

```
└─$ nmap -sV --script vulners --script-args mincvss=7.0 10.80.80.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-20 09:52 EDT
Nmap scan report for 10.80.80.100
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version    port/proto  service
|_  100000    2              111/tcp    rpcbind
|_  100000    2              111/udp    rpcbind
|_  100003    2,3,4          2049/tcp   nfs
|_  100003    2,3,4          2049/udp   nfs
|_  100005    1,2,3          42373/tcp  mountd
|_  100005    1,2,3          56098/udp  mountd
|_  100021    1,3,4          34206/udp  nlockmgr
|_  100021    1,3,4          44137/tcp  nlockmgr
|_  100024    1              56579/udp  status
|_  100024    1              60810/tcp  status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
```

REPORT

A security assessment was conducted on metasploitable2 machine using Kali Linux in lab environment. The assessment involved network reconnaissance and vulnerability analysis using nmap SYN scanning(-sS), service version detection (-sV), and NSE scripts for deeper inspection and vulnerability identification.

The scan revealed multiple open services, including FTP (21), SSH (22), Telnet (23), SMTP (25), HTTP (80) among others. Many of these services were outdated or insecure, exposing the system to multiple attacks.

HIGH SEVERITY FINDINGS

1. Vsftpd 2.3.4 Backdoor (CVE-2011-2523)

This service contains a known backdoor that allows remote attackers to gain shell access.

REMEDIATION: i. Upgrade to a secure and suspected FTP server version.

ii. Remove FTP service if not required.

iii. Replace FTP with secure protocols like FTPS.

IV. Restrict service access using firewall rules.

MEDIUM SEVERITY FINDINGS

1. Telnet Services Exposure (Port 23)

This transmits data in plaintext, making credentials vulnerable to interception

REMEDIATION: i. Disable Telnet completely.

ii. Replace with SSH for secure remote access

iii. Enforce encrypted communication for all remote sessions.

LOW SEVERITY FINDINGS

1. HTTP Server Information Disclosure (Port80)

Web server headers exposure version information useful for attackers

REMEDIATION: i. Disable or modify server banners.

ii. Remove unnecessary web modules.

iii. Apply regular updates and patches.

CONCLUSION

The assessment highlights multiple security weaknesses in metasploitable 2, ranging from critical remote code execution vulnerabilities to information disclosure issues. These findings were identified using Nmap scanning techniques and NSE scripts of which a snap of few out of many was done and attached to this report.

Overall, the system demonstrates how outdated services and poor configuration can significantly increase attack exposure. Implementing proper patch management, disabling unnecessary services, enforcing strong authentication, and using secure communication protocols are essential to reduce risk and improve overall system security.

